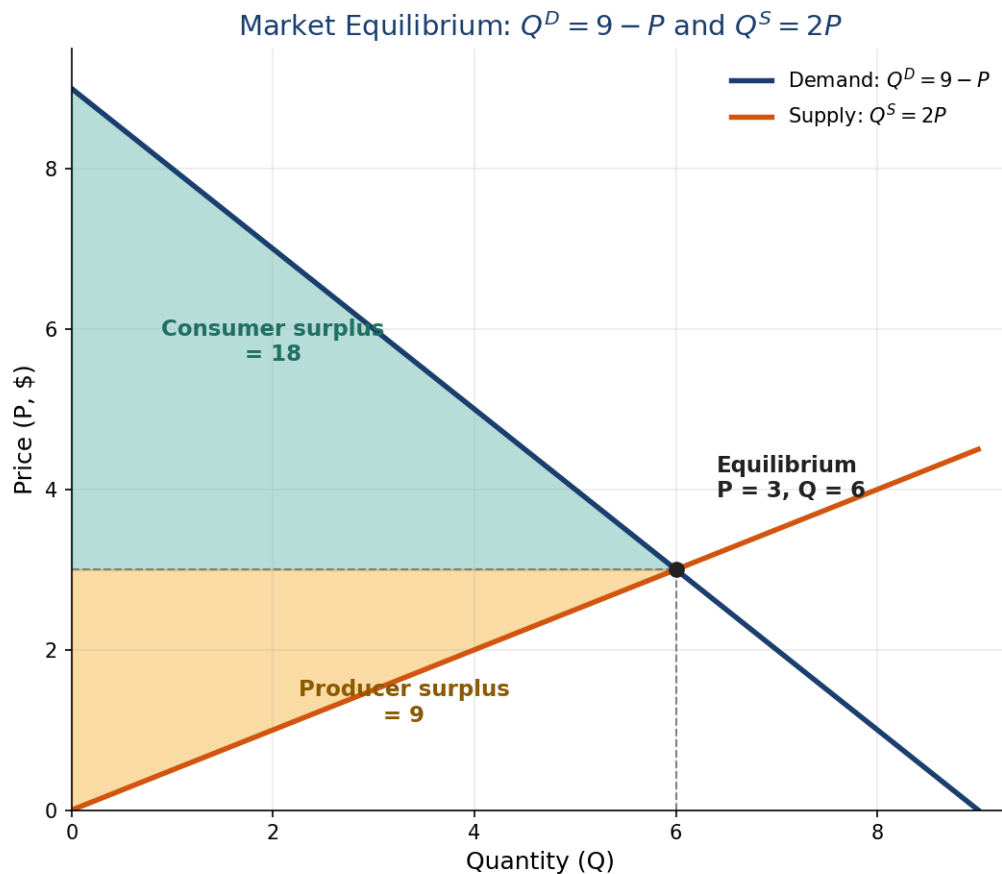


Exercise on Equilibrium- Answers

1. Suppose the following demand function: $Q^D=9-p$. In addition, suppose the following supply function: $Q^S=2p$ where Q^D is the quantity demanded of a good, Q^S is the supplied quantity of a good and p is the price of a good in \$.
 - a. What would be the quantity demanded if the price is 1 \$?
 - b. What would be the quantity demanded if the price is 8 \$?
 - c. What would be the quantity supplied if the price is 1 \$?
 - d. What would be the quantity supplied if the price is 8 \$?
 - e. What would be the price and the quantity in the equilibrium?
 - f. Show it on the graph
 - g. What is the producer surplus?
 - h. What is consumer surplus?
 - i. What is the total **social welfare**?

Answer:

- a. $Q^D=9-1=8$
- b. $Q^D=9-8=1$
- c. $Q^S=2*1=2$
- d. $Q^S=2*8=16$
- e. In equilibrium the supply is equal to demand ($Q^S = Q^D$) therefore, $9-p=2p$. Solving this equation we get $p=3$ and $Q=6$.
- f. See the answer below



- g. The consumer surplus is the area between the demand curve and the price (triangle area) $(9-3) \cdot (6-0) / 2 = 18$
- h. The producer surplus is the area between the supply curve and the price (triangle area) $(3-0) \cdot (6-0) / 2 = 9$
- i. The total social welfare is defined as the sum of consumer and producer surplus.

The total social welfare is therefore $18+9=27$